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Honours Thesis

Meta-Model For Regime-Adaptive Deep Hedging Under Stochastic Volatility

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Summary. This thesis studies the problem of dynamic hedging for options under stochastic volatility, where market incompleteness and regime-dependent behaviour limit the effectiveness of classical approaches such as Black–Scholes delta hedging. The objective is to evaluate whether combining specialised learning-based hedging policies with a regime identification mechanism can improve performance across heterogeneous market environments.

The methodology integrates stochastic modelling with machine learning. Synthetic market data is generated using the Heston stochastic volatility model to simulate distinct volatility regimes. Deep hedging policies are trained separately under low-, medium-, and high-volatility environments, allowing each model to specialise in a specific regime. A meta-classification model is developed to infer the prevailing market regime from observable features such as realised volatility and historical returns, and to select the corresponding hedging policy. The framework is evaluated on both synthetic data and real market data, including periods of elevated volatility.

The results show that the proposed meta-model consistently outperforms classical benchmark strategies overall, while learning-based hedging approaches demonstrate strong performance across both synthetic and real market settings. The findings suggest that incorporating regime awareness can improve robustness under stochastic volatility and model uncertainty.

Statement of Contributions. The main contributions of this thesis are as follows. First, the formulation of regime-adaptive hedging as a supervised learning problem, separating regime identification from policy learning (Sections 2.4 and 3.4). Second, the design and implementation of a meta-model that selects among specialised deep hedging policies based on observable market features (Sections 2.4 and 3.4). Third, the development of a complete computational pipeline, including stochastic market simulation under the Heston model, neural network-based hedging models, and evaluation under both frictionless and transaction cost settings (Sections 2.2, 2.3, 3.3 – 3.6). All modelling, experiments, and analysis are the author's own work.

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Chapter 1

Introduction

1.1 Motivation

Options are among the most widely used financial derivatives for risk transfer, portfolio protection, and volatility trading in modern financial markets [2, 3]. For institutions that sell options, a central task is to dynamically hedge the risk exposure generated by the derivative contract. This involves constructing a trading strategy in the underlying asset that offsets changes in the value of the option over time.

The classical framework for option pricing and hedging was established by the seminal work of Black and Scholes (1973) and Merton (1973). Under a set of idealised assumptions – including frictionless markets, continuous trading, and constant volatility – the Black–Scholes–Merton model demonstrates that a derivative can be perfectly replicated by dynamically trading the underlying asset and a risk-free asset [4, 5]. In this setting, the optimal hedging strategy is given by the well-known delta hedge, which adjusts the position in the underlying according to the sensitivity of the option value to movements in the underlying price [6, 7].

While the Black–Scholes model remains foundational in quantitative finance, extensive empirical evidence demonstrates that its assumptions are violated in real financial markets. In particular, volatility is neither constant nor deterministic. Instead, financial time series exhibit volatility clustering, heavy tails, leverage effects, and time-varying conditional variance [8–11]. During periods of market stress, volatility can increase dra-

matically within short time intervals, as observed during events such as the global financial crisis in 2008 [12, 13].

These empirical observations have motivated the development of stochastic volatility models that explicitly treat volatility as a dynamic process. One of the most influential models in this class is the Heston stochastic volatility model, which introduces a mean-reverting variance process correlated with asset returns [14]. In such models, markets become incomplete because volatility risk cannot generally be perfectly hedged using the underlying asset alone [15, 16].

Consequently, the objective of hedging shifts from perfect replication to minimising hedging error under uncertainty. This raises an important practical question: how can hedging strategies remain effective when market volatility evolves unpredictably and may vary across different market environments?

1.2 Current Obstacles

Hedging under stochastic volatility remains challenging in practice for several reasons.

1.2.1 Incomplete Market Dynamics and Model Uncertainty

Volatility is latent and unobservable. Traders observe asset prices but not the instantaneous volatility process driving price dynamics. As a result, hedging strategies must rely on proxies such as historical volatility, realised volatility, or implied volatility from option markets [17, 18]. These proxies introduce estimation error that can significantly affect hedging performance.

There is also substantial model uncertainty. Even when stochastic volatility models such as the Heston model are used, the parameters governing the volatility process must be estimated from data and may vary across time and assets [18, 19]. A hedging strategy calibrated under one parameter configuration may therefore perform poorly if the true market dynamics differ from those assumed during calibration.

1.2.2 Regime-Dependent Behaviour of Markets

Financial markets frequently exhibit regime-dependent behaviour. Periods of relatively stable volatility can alternate with episodes of extreme turbulence triggered by macroeconomic shocks, geopolitical events, or systemic crises. For example, the COVID-19 pandemic caused a sudden and dramatic increase in market volatility in early 2020. Empirical studies have shown that such regime shifts can fundamentally alter the statistical properties of financial returns [20–22].

These regime changes pose a significant challenge for hedging strategies calibrated under a single market environment. A strategy that performs well during stable periods may deteriorate substantially during episodes of elevated volatility.

1.2.3 Computational Feasibility and Robustness

Finally, hedging strategies must remain computationally feasible and robust. Sophisticated models may achieve lower theoretical hedging error but can be difficult to implement in practice, particularly when they require frequent recalibration or rely on parameters that are difficult to estimate accurately.

These challenges suggest that effective hedging strategies must not only perform well under a particular model specification but also remain robust across heterogeneous market environments.

1.3 Current Approaches

Researchers have developed several approaches to address these challenges.

1.3.1 Classical Approaches

ARCH and GARCH models capture time-varying conditional volatility in financial returns [9, 10], while realised volatility methods use high-frequency data to construct empirical volatility measures [17, 23, 24]. These methods provide improved volatility inputs

for classical delta hedging strategies but still rely on modelling assumptions that may break down during extreme market conditions.

Stochastic volatility models have also been incorporated into pricing and hedging frameworks. The Heston model [14] is widely used in both academic research and industry practice because it captures several empirical features of asset returns while remaining analytically tractable [13, 19]. In incomplete markets, hedging strategies are often formulated using criteria such as mean-variance hedging or local risk minimisation [16, 25, 26].

1.3.2 Learning-Based Approaches

Machine learning techniques have recently been applied to the hedging problem. The deep hedging framework introduced by Buehler et al. (2019) formulates hedging as a sequential decision problem in which a neural network learns a trading policy that minimises a risk objective over simulated market trajectories [27]. By learning directly from data, deep hedging methods can adapt to nonlinear relationships between observable state variables and optimal trading decisions.

Subsequent research has explored reinforcement learning and neural-network-based hedging strategies in increasingly complex environments, including markets with stochastic volatility and transaction costs [28, 29]. These methods often outperform classical delta hedging when evaluated in simulated market environments.

However, most deep hedging models are trained under a single market environment, implicitly assuming that the training distribution matches the deployment environment. In practice, financial markets transition between different volatility regimes, raising the question of whether multiple specialised hedgers may perform better than a single universal policy.

1.4 Literature Review

The foundations of modern derivative pricing were established by Black and Scholes (1973) and Merton (1973), whose work introduced the concept of dynamic replication and the Black–Scholes partial differential equation [4, 5]. Their framework laid the groundwork for the widespread use of delta hedging in financial markets [2, 6].

Empirical research subsequently demonstrated that asset returns exhibit time-varying volatility and other deviations from the assumptions of geometric Brownian motion [8, 11]. ARCH and GARCH models were developed to capture conditional heteroscedasticity in financial time series [9, 10], while realised volatility methods provided direct empirical measures of return variation [17].

Stochastic volatility models such as the Heston model further improved the realism of asset price modelling by introducing a dynamic variance process [14]. These models have been widely used to capture volatility smiles and leverage effects observed in option markets [13].

In incomplete markets, hedging strategies are often formulated using quadratic hedging frameworks such as mean-variance hedging and local risk minimisation [16, 25]. These approaches provide rigorous theoretical foundations but can become computationally complex in high-dimensional settings.

Recent advances in machine learning have led to the development of deep hedging, which uses neural networks to learn hedging policies directly from simulated market data [27]. Follow-up work has further explored reinforcement learning approaches to hedging and portfolio optimisation [28, 29].

Finally, regime-switching models have been widely used in econometrics to capture structural changes in financial time series [20, 21]. In machine learning, mixture-of-experts models combine multiple specialised predictors with a gating network that selects among them based on the input data [30, 31].

Despite these advances, deep hedging approaches are often trained under a single market environment, which may limit their robustness under changing volatility conditions.

This suggests the need for frameworks that can adapt to heterogeneous market regimes.

Motivated by these ideas, this thesis proposes a regime-adaptive deep hedging framework that combines specialised hedging policies with a meta-classification model capable of selecting the most appropriate hedger based on observable market features. The following chapter introduces the theoretical framework and modelling assumptions underlying this approach.

Chapter 2

Methodology

This chapter introduces the theoretical and computational framework used to study regime-adaptive option hedging. The objective is to evaluate whether specialised hedging policies trained under different volatility environments can improve robustness across heterogeneous market conditions.

The methodology combines classical financial modelling with machine learning techniques. Synthetic market environments are first generated using stochastic volatility models, allowing the construction of controlled training datasets that capture different volatility regimes. Hedging strategies are then implemented using both traditional analytical approaches and neural-network-based policies trained on simulated data. Finally, a meta-classification model is developed to infer the prevailing market regime using observable features derived from historical returns and realised volatility.

This framework integrates ideas from quantitative finance, stochastic modelling, and machine learning. From the financial perspective, the problem is framed as a dynamic hedging task in incomplete markets. From a data science perspective, the approach can be viewed as a supervised learning pipeline in which simulated market trajectories provide training data for both hedging policies and regime classification models.

2.1 Theoretical Foundations

2.1.1 Hedging as a Risk Minimisation Problem

Consider a financial market consisting of an underlying asset with price process S_t and a derivative with payoff $H(S_T)$ at maturity T. A trader who sells the derivative must construct a trading strategy that hedges the resulting exposure.

In the classical Black–Scholes framework, the derivative payoff can be perfectly replicated through continuous trading in the underlying asset and a risk-free asset. However, when volatility is stochastic or trading occurs at discrete intervals, perfect replication is generally not achievable. In such settings, the hedging problem is often formulated as a risk minimisation task.

Let V_T denote the terminal value of a hedging portfolio constructed using a trading strategy ϕ_t. The hedging objective can be expressed as minimising the expected squared replication error:

$$\min_{\phi} \mathbb{E}\left[(V_T - H(S_T))^2\right]. \tag{2.1}$$

This formulation corresponds to the mean-variance hedging framework commonly used in incomplete markets [16]. The goal is not perfect replication but rather the construction of a strategy that minimises the expected deviation between the portfolio value and the option payoff.

2.1.2 Market Incompleteness under Stochastic Volatility

In models with stochastic volatility, the asset price dynamics depend on an additional state variable representing the volatility process. Since volatility cannot generally be traded directly, the number of risk factors exceeds the number of tradable assets, resulting in an incomplete market.

As a consequence, multiple hedging strategies may exist that partially hedge the derivative exposure, but none can eliminate risk entirely. Hedging performance therefore

depends on how effectively the chosen strategy responds to changes in both asset prices and volatility dynamics.

2.1.3 Discrete-Time Hedging Framework

In practice, hedging strategies are implemented at discrete time intervals. Let $t_0, t_1, \ldots, t_T$ denote a sequence of trading times. At each step, the hedger selects a position ϕ_t in the underlying asset.

The value of the hedging portfolio evolves according to

$$V_{t+1} = V_t + \phi_t(S_{t+1} - S_t). \tag{2.2}$$

The performance of a hedging strategy therefore depends on how the position ϕ_t is chosen as a function of observable information, such as current prices, historical returns, and estimated volatility measures.

This formulation naturally lends itself to learning-based methods in which the mapping from observable state variables to hedging decisions is learned from simulated or historical market data.

2.2 Market Models and Data Generation

To evaluate hedging strategies across different volatility environments, it is necessary to generate market trajectories under controlled stochastic models. Synthetic data generation plays a central role in this study because it allows the construction of large datasets covering a range of volatility regimes that may not be easily observed in historical data alone.

The approach adopted in this thesis follows a simulation-based learning framework. First, asset price paths are generated using stochastic models of financial markets. These simulated trajectories are then used to construct datasets consisting of asset prices, returns, and derived volatility features. Hedging strategies are subsequently evaluated by applying trading policies along these simulated market paths.

Two market models are considered. The first is the geometric Brownian motion model, which represents the classical Black–Scholes setting with constant volatility. The second is the Heston stochastic volatility model, which introduces time-varying volatility and captures several empirical properties of financial markets.

2.2.1 Geometric Brownian Motion

The geometric Brownian motion (GBM) model provides the foundation for the classical Black–Scholes framework. Under GBM dynamics, the asset price process follows the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \tag{2.3}$$

where μ represents the drift, σ denotes the constant volatility parameter, and W_t is a standard Brownian motion.

This model assumes that volatility remains constant over time and that asset returns are normally distributed. While these assumptions are known to be violated in real financial markets, GBM remains an important benchmark because it leads to closed-form option pricing formulas and provides a baseline environment for evaluating hedging strategies.

In discrete time, the GBM process can be simulated using the log-Euler discretisation

$$S_{t+1} = S_t \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right), \tag{2.4}$$

where $Z_t \sim \mathcal{N}(0,1)$ and Δt denotes the time step. This discretisation ensures that simulated prices remain strictly positive.

Although GBM is overly simplistic for modelling real markets, it provides a useful reference environment in which classical delta hedging strategies are theoretically optimal.

2.2.2 Heston Stochastic Volatility Model

To capture time-varying volatility, this study adopts the Heston stochastic volatility model [14]. In this framework, both the asset price and the instantaneous variance evolve as stochastic processes:

$$dS_t = \mu S_t dt + \sqrt{v_t}\, S_t dW_t^{(1)}, \tag{2.5}$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t} dW_t^{(2)}. \tag{2.6}$$

Here v_t represents the instantaneous variance of the asset return process. The parameter κ controls the speed of mean reversion toward the long-run variance level θ, while ξ governs the volatility of volatility. The two Brownian motions $W_t^{(1)}$ and $W_t^{(2)}$ are correlated with correlation coefficient ρ.

The Heston model captures several stylised facts observed in financial markets. In particular, it produces stochastic volatility, volatility clustering, and leverage effects arising from the correlation between price shocks and volatility shocks.

Unlike the Black–Scholes model, markets under stochastic volatility are generally incomplete because volatility risk cannot be perfectly hedged using the underlying asset alone. As a result, hedging strategies cannot eliminate risk entirely and must instead minimise hedging error under uncertainty.

2.2.3 Monte Carlo Market Simulation

Because closed-form solutions for hedging strategies under stochastic volatility are generally unavailable, simulation-based approaches are widely used to evaluate hedging performance.

In this study, Monte Carlo simulation is used to generate large numbers of synthetic market trajectories under the Heston model. Each simulated trajectory represents a potential evolution of asset prices and volatility over a fixed time horizon.

To ensure numerical stability, the variance process is simulated using the full truncation Euler scheme, which prevents negative variance values during discretisation. The asset price process is then updated using a log-Euler discretisation conditional on the simulated variance.

2.3 Hedging Policies

This section introduces the hedging strategies evaluated in this study. The objective is to compare classical analytical hedging approaches with learning-based strategies learned from simulated market data.

Three categories of hedging policies are considered. The first is the classical Black–Scholes delta hedge, which serves as a benchmark under the assumption of constant volatility. The second extends this approach by incorporating time-varying volatility estimates derived from historical data. The third approach uses neural networks to learn hedging policies directly from simulated market trajectories, following the deep hedging framework proposed by [27].

2.3.1 Black–Scholes Delta Hedging

Under the Black–Scholes framework, the value of a European call option with strike price K and maturity T is given by

$$C(S_t, t) = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2), \tag{2.7}$$

where

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \tag{2.8}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}, \tag{2.9}$$

and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distri-

bution.

The optimal hedging position in the underlying asset is given by the option's delta:

$$\Delta_t = \frac{\partial C}{\partial S_t} = \Phi(d_1). \tag{2.10}$$

In practice, delta hedging involves adjusting the position in the underlying asset at each rebalancing time step according to this sensitivity. When the model assumptions hold and trading occurs continuously, this strategy replicates the option payoff exactly.

However, when volatility is stochastic or trading occurs discretely, the Black–Scholes delta hedge becomes imperfect and may generate substantial hedging error.

2.3.2 Volatility-Adjusted Delta Hedging

To partially address the limitation of constant volatility in the Black–Scholes model, volatility estimates can be updated dynamically using historical return data.

One common approach is to estimate volatility using exponentially weighted moving averages (EWMA). Let r_t denote the log return at time t. The EWMA variance estimate evolves according to

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \tag{2.11}$$

where $\lambda \in (0, 1)$ is a decay parameter that controls the persistence of volatility estimates.

This estimated volatility is then substituted into the Black–Scholes formula to compute the delta used for hedging. The resulting strategy adapts to recent market conditions by increasing hedge sensitivity during periods of elevated volatility and reducing it during calmer market environments.

Although this approach improves responsiveness to changing market conditions, it still relies on the Black–Scholes structural assumptions and may fail to capture complex nonlinear relationships between market variables and optimal hedging decisions.

2.3.3 Deep Hedging

Deep hedging approaches treat the hedging problem as a sequential decision-making task in which the hedger selects trading positions based on observable market information.

Following the framework introduced by [27], a neural network is used to parameterise the hedging policy. At each time step t, the policy takes observable market variables as input and outputs the position ϕ_t in the underlying asset:

$$\phi_t = f_\theta(x_t), \tag{2.12}$$

where f_θ denotes a neural network with parameters θ, and x_t represents the set of observable state variables, such as log-moneyness and time-to-maturity.

The neural network parameters are learned by minimising a loss function defined over simulated market trajectories. In this study, the objective is to minimise the mean squared hedging error between the terminal portfolio value and the option payoff:

$$\min_\theta \ \mathbb{E}\left[(V_T - H(S_T))^2\right]. \tag{2.13}$$

Because the policy is trained on large numbers of simulated market paths, the neural network can learn complex nonlinear relationships between market states and optimal hedging decisions that may not be captured by analytical models.

In this study, separate deep hedging policies are trained under different stochastic volatility environments corresponding to low, medium, and high volatility regimes. These specialised hedgers are later combined using a meta-model that selects the most appropriate policy based on observable market features.

While the above formulation assumes frictionless trading, practical hedging strategies must account for transaction costs arising from rebalancing. The following subsection incorporates such trading frictions into the hedging framework.

2.3.4 Hedging with Transaction Costs

The baseline hedging formulations considered thus far assume frictionless trading. However, in practical settings, each portfolio rebalancing incurs transaction costs, which can significantly impact hedging performance.

To account for this, a transaction cost model reflecting brokerage pricing for fractional share trading is introduced. At each rebalancing step t, the traded notional is given by:

$$\text{TradeValue}_t = S_t \cdot |\phi_t - \phi_{t-1}|, \tag{2.14}$$

where S_t is the asset price and ϕ_t denotes the position in the underlying asset at time t. The corresponding transaction cost is defined as:

$$\text{tc}_t = \max\left(c_{\min},\ c \cdot \text{TradeValue}_t\right), \tag{2.15}$$

where $c > 0$ is the proportional transaction cost rate and $c_{\min} > 0$ represents a minimum fee per trade. This formulation captures the fee structure commonly applied in practice, where small trades are dominated by a fixed minimum cost [32].

Under this specification, the self-financing portfolio dynamics are modified accordingly. Let V_t denote the portfolio value at time t. The wealth recursion becomes:

$$V_{t+1} = \left(V_t - \phi_t S_t - \text{tc}_t\right) e^{r\Delta t} + \phi_t S_{t+1}. \tag{2.16}$$

This adjustment is applied uniformly across all hedging strategies considered in this study, including both Black–Scholes delta hedging and deep hedging policies.

The inclusion of transaction costs introduces a trade-off between hedging accuracy and trading activity. Frequent rebalancing reduces replication error but increases cumulative trading costs, while infrequent adjustments lower costs at the expense of greater exposure to price fluctuations.

Within the deep hedging framework, transaction costs are incorporated directly into the portfolio dynamics used during training, allowing the neural network to learn cost-

aware hedging strategies. In particular, the presence of a minimum fee introduces a nonlinearity in trading costs, encouraging the model to avoid small, frequent adjustments and instead adopt more selective rebalancing behaviour. Transaction costs are also consistently accounted for during evaluation to ensure a fair comparison across all hedging methods.

2.4 Meta-Model for Regime-Adaptive Hedging

The deep hedging policies introduced in the previous section are trained under specific stochastic volatility environments. Each model therefore specialises in a particular volatility regime. While such specialised models may perform well when the market environment matches the training regime, their performance may deteriorate when applied to markets with different volatility characteristics.

In practice, the prevailing volatility regime is not directly observable. Instead, it must be inferred from observable market data such as recent returns and realised volatility measures. This motivates the development of a meta-model capable of selecting the most appropriate hedging policy based on observable features of the market state.

From a machine learning perspective, this problem can be formulated as a classification task. Given a set of observable features describing recent market behaviour, the objective is to predict which volatility regime is most consistent with the current environment. The predicted regime is then used to select the corresponding specialised hedging policy.

2.4.1 Regime Identification as a Learning Problem

Let x denote a vector of observable market features constructed from historical asset returns. The meta-model seeks to learn a mapping

$$g(x) \to \{1, 2, \ldots, K\}, \tag{2.17}$$

where each class corresponds to a volatility regime associated with a specialised deep

hedging policy.

In this study, three volatility regimes are considered: low volatility, medium volatility, and high volatility. Each regime corresponds to a distinct set of parameters in the stochastic volatility model used to generate simulated training data.

The classification model is trained using labelled datasets constructed from simulated market trajectories. For each simulated episode, the true volatility regime is known because the market parameters used for simulation are controlled by the experiment design. These regime labels therefore provide ground truth for supervised learning.

2.4.2 Feature Engineering

To infer volatility regimes from observable market data, a set of features is constructed from historical returns available prior to the hedging window. These features summarise recent market behaviour and provide inputs to the classification model.

Let $r_t = \log(S_t/S_{t-1})$ denote the log return at time t. Several feature groups are computed from historical returns:

- **Realised volatility measures:** Realised volatility over multiple rolling windows provides estimates of recent return variability. For a window length w, realised volatility is computed as

$$RV_w = \sqrt{252} \cdot \mathrm{Std}(r_{t-w+1}, \ldots, r_t). \tag{2.18}$$

- **Absolute return statistics:** The mean and standard deviation of absolute returns over recent windows provide additional information about return dispersion.

- **Recent return observations:** Lagged returns over short horizons capture short-term market momentum and recent price shocks.

- **Current price level:** The asset price at the start of the hedging window is included as an additional contextual variable.

These features collectively provide a compact representation of recent market conditions that can be used to infer the underlying volatility regime.

2.4.3 Meta-Classification Model

The regime prediction task is implemented using a supervised classification model. Given the feature vector x, the classifier estimates the probability that the current market state belongs to each volatility regime.

Formally, the classifier estimates

$$P(y = k \mid x), \quad k \in \{1, 2, 3\}, \tag{2.19}$$

where y denotes the regime label.

The model is trained on datasets generated from simulated market trajectories. Each episode is labelled according to the volatility regime used during simulation. Standard supervised learning techniques are then used to learn the mapping between feature vectors and regime labels.

Once trained, the classifier can be applied to previously unseen market data to predict the most likely volatility regime at the start of each hedging episode.

2.4.4 Policy Selection Framework

The meta-model integrates the regime classifier with the specialised deep hedging policies trained in earlier sections. The overall framework operates as follows:

1. At the start of each hedging episode, observable features are computed from historical returns.

2. The feature vector is fed into the regime classification model to predict the prevailing volatility regime.

3. The deep hedging policy corresponding to the predicted regime is selected.

4. The selected policy is applied to determine the hedge position throughout the episode.

This approach can be viewed as a mixture-of-experts framework in which multiple specialised models are combined through a gating mechanism that selects the most appropriate expert based on observable inputs.

By separating regime identification from policy learning, the framework aims to improve robustness across heterogeneous market environments while retaining the performance advantages of specialised deep hedging policies.

From a decision-theoretic perspective, this framework can be interpreted as approximating an oracle policy that selects the optimal hedging strategy conditional on the underlying market regime. In practice, the true regime is unobservable, and the meta-classification model acts as an estimator that maps observable features to an appropriate policy.

This decomposition allows the model to adapt to heterogeneous environments by selecting among specialised hedgers, thereby reducing the impact of model misspecification that may arise when a single policy is applied across different market conditions.

Chapter 3

Experiments and Results

This chapter presents the experimental framework and empirical results used to evaluate the regime-adaptive hedging approach proposed in this thesis. The objective of the experiments is to assess whether specialised deep hedging policies, combined with a meta-classification model for regime selection, can improve hedging performance across heterogeneous volatility environments.

The evaluation proceeds in two stages. First, controlled experiments are conducted using synthetic market data generated under stochastic volatility models. These simulations allow the performance of different hedging strategies to be compared under known volatility regimes. In this setting, the true market parameters are known, enabling systematic evaluation of regime-specific hedging policies and the proposed meta-model.

Second, the trained models are applied to historical equity market data in order to examine how the framework performs under real market conditions. The empirical evaluation focuses on the period surrounding the COVID-19 market shock, which provides a natural setting to study hedging behaviour under rapidly changing volatility environments.

The remainder of this chapter describes the experimental design, parameter configurations, training procedures, and evaluation metrics used in the study, followed by the results obtained from both synthetic simulations and real market data.

The experiments are designed to ensure reproducibility and controlled comparison between hedging strategies by keeping trading assumptions, hedging horizons, and option

specifications consistent across all evaluations.

3.1 Experimental Design

The objective of the experiments is to evaluate the hedging performance of different trading strategies under varying market environments. In particular, the experiments investigate whether regime-adaptive hedging policies can improve robustness when market volatility changes across time.

Four classes of hedging strategies are compared:

- Black–Scholes delta hedging with constant volatility,
- volatility-adjusted delta hedging using exponentially weighted moving average (EWMA) volatility estimates,
- specialised deep hedging policies trained under specific volatility regimes,
- a meta-model that selects among specialised deep hedging policies based on observable market features.

All strategies are evaluated under identical trading assumptions in order to ensure fair comparison. The experiments consider European call options written on the underlying asset. Hedging is performed through dynamic trading in the underlying asset over a fixed time horizon.

Each experiment is organised into episodes of length 40 trading days. The first 20 days of each episode are used to construct observable features from historical returns. These features are used either for volatility estimation in classical models or as inputs to the meta-classification model. The remaining 20 days constitute the hedging window during which trading decisions are executed.

At the beginning of the hedging window, the option is priced using a Monte Carlo estimator consistent with the underlying stochastic volatility model. The hedging strategy then determines the position in the underlying asset at each rebalancing step. At the end

of the episode, the replication error between the terminal portfolio value and the option payoff is recorded.

By structuring the experiments in this manner, all hedging strategies are evaluated under the same market trajectories, allowing direct comparison of their replication performance.

The following sections describe the parameter configurations used to generate synthetic market environments, the training procedure for the deep hedging policies and meta-classification model, and the evaluation metrics used to assess hedging performance.

3.2 Market Parameter Settings

Synthetic market data used in the experiments is generated using the Heston stochastic volatility model described in Chapter 2. The model parameters are chosen to represent a range of volatility regimes commonly observed in financial markets.

The Heston model is characterised by five parameters: the mean-reversion rate κ, the long-run variance θ, the volatility of volatility ξ, the correlation ρ between price and volatility shocks, and the initial variance v_0. By varying these parameters, it is possible to construct market environments exhibiting different volatility dynamics.

In this study, six parameter configurations are used to generate synthetic market trajectories. These configurations are grouped into three volatility regimes: low volatility, medium volatility, and high volatility. Each regime is represented by two parameter sets with different volatility-of-volatility values in order to introduce additional variability within each regime.

The parameter settings used in the experiments are summarised in Table 3.1.

These parameter configurations generate market environments with progressively increasing volatility levels. The long-run variance parameter θ determines the baseline volatility level for each regime, while the volatility-of-volatility parameter ξ controls the variability of the variance process.

The negative correlation parameter $\rho = -0.7$ captures the leverage effect commonly

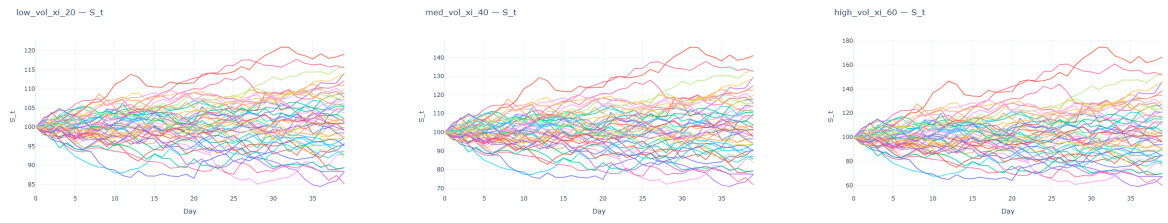
Regime	κ	θ	ξ	ρ	v_0
Low volatility (1)	2.0	0.04	0.20	-0.7	0.04
Low volatility (2)	2.0	0.04	0.38	-0.7	0.04
Medium volatility (1)	2.0	0.16	0.40	-0.7	0.16
Medium volatility (2)	2.0	0.16	0.75	-0.7	0.16
High volatility (1)	2.0	0.36	0.60	-0.7	0.36
High volatility (2)	2.0	0.36	1.10	-0.7	0.36

Table 3.1: Heston parameter configurations used to generate synthetic market environments.

observed in equity markets, where negative price shocks tend to coincide with increases in volatility.

For each parameter configuration, multiple price trajectories are simulated using Monte Carlo methods. Each trajectory forms a 40-day episode, consisting of a 20-day feature window followed by a 20-day hedging period as described in Section 3.1. These simulated episodes are used to train and evaluate the deep hedging policies and the meta-classification model.

Figure 3.1 illustrates representative simulated price paths under the three broad volatility regimes. As expected, paths generated under the low-volatility regime evolve more smoothly, while those under the high-volatility regime exhibit substantially greater fluctuation and dispersion. These differences motivate the use of regime-specific hedging policies, since a strategy trained under one volatility environment may not generalise well to another.



(a) Low volatility (b) Medium volatility (c) High volatility

Figure 3.1: Representative simulated price paths under different Heston volatility regimes.

3.3 Training of Deep Hedging Policies

The deep hedging policies are trained using simulated market trajectories generated under the Heston stochastic volatility model described in Section 2.2.2. Each neural network learns a mapping from observable market features to hedge positions in the underlying asset, with the objective of minimising hedging error over a fixed hedging horizon.

To account for heterogeneous volatility environments, separate deep hedging models are trained for different volatility regimes. These specialised hedgers are later used as candidate policies within the meta-model framework introduced in Section 2.4.

3.3.1 Training Data Generation

Training data for the deep hedgers is generated using Monte Carlo simulation of the Heston stochastic volatility model. Each simulation produces a full market episode consisting of 40 trading days.

Within each episode, hedging decisions are applied over the final 20 days. The first 20 days provide historical information used to construct observable market features.

For each volatility regime, a large number of independent market episodes are simulated using the corresponding Heston parameters described in Table 3.1. Because the simulation environment can generate arbitrarily many trajectories, the training dataset is effectively unbounded, allowing the neural networks to be trained across diverse simulated market conditions.

3.3.2 Neural Network Architecture

Each deep hedging policy is implemented using a feedforward neural network that maps observable state variables to hedge positions in the underlying asset.

At each trading step t, the model receives a feature vector x_t, defined as:

$$x_t = [\log(S_t/K), \tau_t/T] \tag{3.1}$$

where,

- $\log(S_t/K)$ denotes log-moneyness
- τ_t/T denotes normalised time-to-maturity

The neural network represents a function as shown in Equation 2.12.

The architecture consists of three fully connected hidden layers with 128 neurons in each layer. Nonlinear activation functions are applied after each hidden layer to capture nonlinear relationships between market features and hedging decisions.

Parameter	Value
Episode length	40 days
Feature window	20 days
Hedging horizon	20 days
Neural network depth	3 layers
Activation function	ReLU
Output dimension	1 (hedge ratio)
Hidden units per layer	128
Batch size	1024
Training epochs	4000
Learning rate	1e-3
Optimiser	Adam

Table 3.2: Training configuration for deep hedging models.

The same neural network architecture and hyperparameters are used for all regime-specific deep hedgers. Only the simulated training environment differs across regimes.

3.3.3 Training Procedure

The neural networks are trained using stochastic gradient-based optimisation. The objective is to minimise the mean squared hedging error between the terminal portfolio value and the option payoff as shown in Equation 2.13 over simulated market episodes.

Two training settings are considered in this study: a frictionless setting and a setting with proportional transaction costs.

In the frictionless setting, the portfolio dynamics follow the standard self-financing recursion. In the transaction cost setting, the portfolio dynamics are adjusted according to Equation 2.16, where trading costs are incurred at each rebalancing step.

Training is performed using mini-batch gradient descent with a batch size of 1024. Each network is trained for 4000 epochs, allowing the model to converge to a stable hedging policy under the simulated market environment.

Because the training data is generated through Monte Carlo simulation, the effective dataset size is not fixed. Instead, the simulation environment continuously produces new market trajectories during training, exposing the model to a wide range of market conditions.

At each training step, batches of simulated episodes are generated using Monte Carlo sampling. The neural network then updates its parameters based on the hedging error observed along these trajectories. By continuously sampling new price paths, the training process effectively approximates expectations under the stochastic volatility model.

In the transaction cost setting, trading costs are incorporated directly into the portfolio dynamics during training. This allows the neural network to learn hedging strategies that balance replication accuracy against trading costs.

Separate neural networks are trained for each volatility regime defined in Section 3.2. These specialised hedgers capture regime-specific hedging behaviour under both frictionless and cost-aware settings.

3.4 Meta-Model Training

This section describes the construction of the dataset used to train the regime classifier introduced in Section 2.4. The training procedure consists of three main steps: feature construction, oracle label generation, and classifier training.

3.4.1 Feature Construction

For each simulated market episode, a feature vector is constructed using only information available prior to the hedge initiation time t_0, as described in Section 2.4.2.

All features are computed strictly using historical observations to avoid forward-looking information.

Table 3.3: Feature variables used for meta-model training.

Feature	Description
RV5, RV10, RV20	Realised volatility over rolling windows
absret_mean_w	Mean absolute return over window w
absret_std_w	Std of absolute returns over window w
ret_last_k	Lagged return observations
ret_mean_20	Mean return over last 20 steps
ret_std_20	Return volatility over last 20 steps

3.4.2 Oracle Label Generation

The training labels for the meta-classifier are obtained by assigning the deep hedging policy that corresponds to the market regime of the episode.

- Label 0: Deep hedger trained on low-volatility regime
- Label 1: Deep hedger trained on medium-volatility regime
- Label 2: Deep hedger trained on high-volatility regime

3.4.3 Classifier Training

The regime classifier is trained using the feature vectors and oracle labels generated from the simulated market episodes.

Let x_i denote the feature vector associated with episode i. The classifier learns a mapping

$$\hat{y}_i = f_\psi(x_i), \tag{3.2}$$

where f_ψ represents the trained classification model.

The trained classifier is subsequently used during evaluation to select the appropriate specialised hedging policy based on the observable market features at the start of each episode.

3.5 Evaluation on Synthetic Data

The proposed regime-adaptive framework is evaluated on synthetic market trajectories generated under the Heston stochastic volatility model. This setting allows direct comparison between classical hedging strategies, specialised deep hedgers, and the meta-model under controlled volatility regimes.

Two evaluation settings are considered: a frictionless setting and a setting with proportional transaction costs. This enables a systematic analysis of hedging performance both in idealised and more realistic market conditions.

3.5.1 Evaluation Setup

A separate set of simulated market episodes is generated for evaluation using the Heston parameter configurations described in Section 3.2. These episodes are not used during model training.

Each episode consists of 40 trading days. The first 20 days are used to construct observable features, while hedging decisions are applied over the remaining 20 days.

For every episode, the following hedging strategies are evaluated:

- Black–Scholes delta hedging with constant volatility,
- Black–Scholes delta hedging using EWMA volatility estimates,
- specialised deep hedging policies trained for the low-, medium-, and high-volatility regimes,
- the proposed meta-model, which selects among specialised deep hedgers based on observable features.

Hedging performance is measured using the replication error of the terminal hedging portfolio relative to the option payoff. For a set of N episodes with replication errors ϵ_i, performance is summarised using the root mean squared error (RMSE), mean absolute error (MAE), and mean hedging error:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\epsilon_i^2}, \tag{3.3}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\epsilon_i|, \tag{3.4}$$

$$\text{Mean} = \frac{1}{N}\sum_{i=1}^{N}\epsilon_i. \tag{3.5}$$

3.5.2 Frictionless Setting

Per-Regime Results

Table 3.4 reports the hedging performance of the evaluated strategies across the six Heston market environments.

Across all six market configurations, the best-performing specialised deep hedger aligns with the corresponding volatility regime: the low-volatility hedger performs best in low-volatility markets, the medium-volatility hedger performs best in medium-volatility markets, and the high-volatility hedger performs best in high-volatility markets. This confirms that the regime-specific models successfully learn environment-dependent hedging behaviour. The meta model performs competitively against the specialised deep hedgers.

Pooled Results

To evaluate robustness across heterogeneous market conditions, all synthetic episodes are pooled and each hedging strategy is evaluated on the combined dataset. The pooled results are reported in Table 3.5.

Table 3.4: Hedging performance across volatility regimes (50 episodes per market).

Market	**Method**	**RMSE**	**MAE**	**Mean**
Low vol ($\xi = 0.20$)	BS_const	0.00533	0.00441	0.00135
	BS_EWMA	0.00698	0.00503	-0.00365
	Deep_low	**0.00397**	0.00325	0.00090
	Deep_med	0.00654	0.00546	0.00072
	Deep_high	0.00866	0.00727	0.00057
	Meta_model	**0.00397**	0.00325	0.00090
Low vol ($\xi = 0.38$)	BS_const	0.00776	0.00632	0.00182
	BS_EWMA	0.00801	0.00545	-0.00328
	Deep_low	**0.00469**	0.00381	0.00057
	Deep_med	0.00680	0.00568	0.00060
	Deep_high	0.00861	0.00722	0.00045
	Meta_model	0.00470	0.00380	0.00052
Med vol ($\xi = 0.40$)	BS_const	0.01077	0.00893	0.00275
	BS_EWMA	0.01388	0.01008	-0.00725
	Deep_low	0.01294	0.01037	0.00175
	Deep_med	**0.00887**	0.00719	0.00090
	Deep_high	0.01009	0.00821	0.00115
	Meta_model	0.00914	0.00735	0.00095
Med vol ($\xi = 0.75$)	BS_const	0.01540	0.01258	0.00375
	BS_EWMA	0.01580	0.01077	-0.00645
	Deep_low	0.01479	0.01103	0.00087
	Deep_med	**0.00995**	0.00799	0.00059
	Deep_high	0.01086	0.00892	0.00088
	Meta_model	0.01024	0.00831	0.00104
High vol ($\xi = 0.60$)	BS_const	0.01632	0.01351	0.00420
	BS_EWMA	0.02069	0.01509	-0.01081
	Deep_low	0.02625	0.02073	0.00272
	Deep_med	0.01693	0.01367	0.00072
	Deep_high	**0.01301**	0.01079	0.00145
	Meta_model	0.01406	0.01144	0.00091
High vol ($\xi = 1.10$)	BS_const	0.02280	0.01864	0.00573
	BS_EWMA	0.02331	0.01595	-0.00955
	Deep_low	0.02939	0.02207	0.00128
	Deep_med	0.01840	0.01365	0.00029
	Deep_high	**0.01473**	0.01197	0.00081
	Meta_model	0.01525	0.01238	0.00043

Table 3.5: Pooled hedging performance (OOS)

Method	RMSE	MAE	Mean
BS_const	0.01430	0.01073	0.00327
BS_EWMA	0.01595	0.01039	-0.00683
Deep_low	0.01815	0.01188	0.00135
Deep_med	0.01219	0.00894	0.00064
Deep_high	0.01122	0.00906	0.00088
Meta Model	**0.01046**	0.00776	0.00079

The pooled results (Visualised in Figure 3.2) show that the regime-adaptive strategy achieves the lowest overall RMSE and MAE across the combined synthetic dataset. While individual specialised hedgers perform best when the regime is known in advance, the automatic regime-selection framework provides the strongest aggregate performance when market conditions vary across episodes.

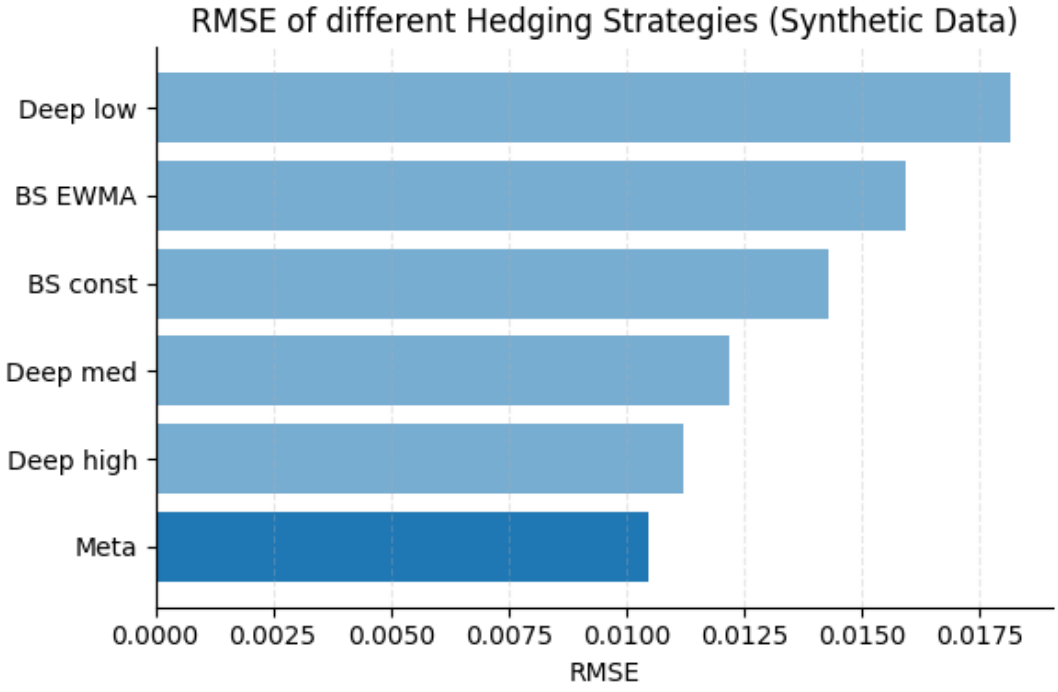


Figure 3.2: RMSE across episodes for different hedging strategies (Synthetic Data)

3.5.3 Setting with Transaction Costs

The same evaluation is repeated under transaction costs, using the cost specification introduced in Equation 2.15. In particular, transaction costs are computed as the maxi-

mum of a proportional fee and a minimum charge, with parameters calibrated to typical brokerage pricing. Specifically, a fee rate of 1% of the traded notional and a minimum fee of USD 0.01 per trade are applied [32]. The pooled results are summarised in Table 3.6.

Table 3.6: Pooled hedging performance with transaction costs.

Method	RMSE	MAE	Mean
BS_const	0.02250	0.01784	-0.01543
BS_EWMA	0.03193	0.02711	-0.02701
Deep_low	0.03056	0.01818	-0.01146
Deep_med	0.02357	0.01499	-0.01380
Deep_high	**0.01939**	**0.01459**	-0.01326
Meta Model	0.01991	0.01464	-0.01325

The inclusion of transaction costs leads to a noticeable deterioration in hedging performance across all methods, reflecting the impact of trading frictions on dynamic rebalancing. This deterioration is also accompanied by uniformly negative mean hedging errors, indicating that transaction costs introduce a systematic downward bias in terminal portfolio value relative to the option payoff.

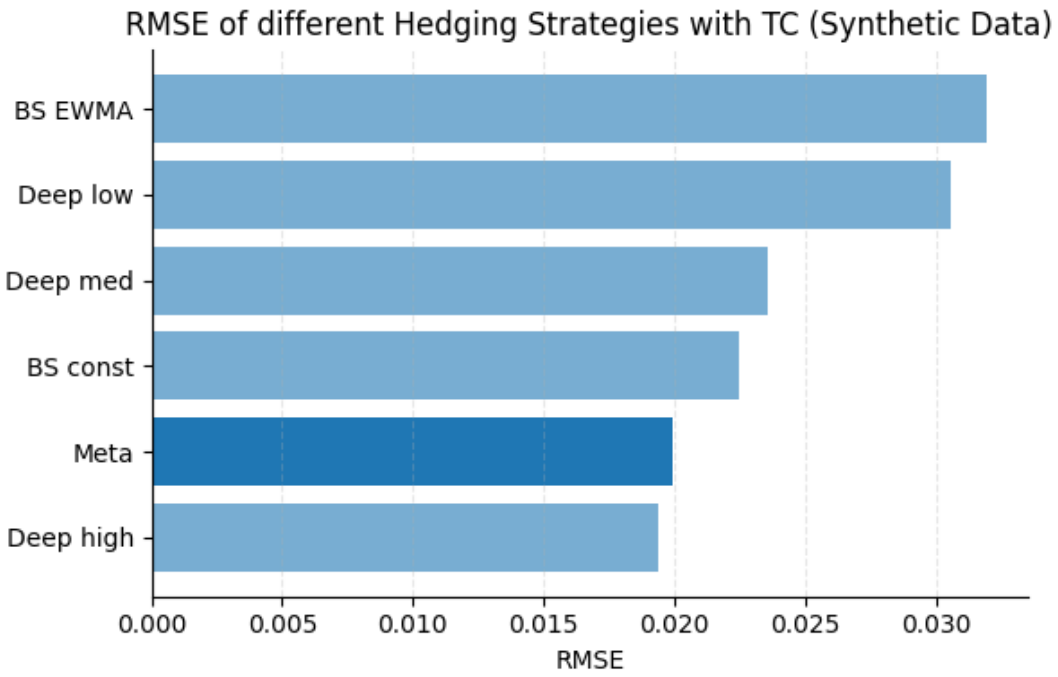


Figure 3.3: RMSE across episodes for different hedging strategies with transaction costs (Synthetic Data)

Among the evaluated strategies, the high-volatility deep hedger achieves the lowest

RMSE and MAE, while the meta-model remains highly competitive and performs only slightly worse. In contrast, the Black–Scholes EWMA strategy deteriorates substantially under transaction costs, suggesting that strategies relying on more reactive hedge adjustments are more heavily penalised when rebalancing is costly.

Overall, these results show that deep hedging policies remain more robust than the classical benchmarks when transaction costs are present. They also suggest that, in the cost-aware setting, a robust specialised policy may outperform the regime-adaptive selector in aggregate, although the meta-model continues to deliver strong overall performance across heterogeneous market environments.

3.6 Real Market Evaluation

To assess the robustness of the proposed regime-adaptive framework outside the simulated training environment, the hedging models are evaluated on historical equity price data. This experiment tests whether policies trained under stochastic volatility simulations can generalise to real financial markets.

3.6.1 Dataset Construction

Daily price data is obtained for a set of large-cap U.S. equities and exchange-traded funds:

JPM, VOO, AMZN, JNJ, AAPL, GOOGL, KO, PG, XOM, META

These assets were selected to provide exposure to multiple sectors and distinct volatility characteristics. The dataset includes a broad market index (VOO), financials (JPM), healthcare (JNJ), consumer staples (KO, PG), energy (XOM), and large-cap technology and growth firms (AAPL, AMZN, GOOGL, META). This broad sector coverage allows the evaluation to capture a range of market dynamics under a common historical period.

The sample period spans

1 September 2019 – 31 May 2021

This period includes the extreme volatility associated with the COVID-19 market crash in early 2020, providing a challenging out-of-sample environment for evaluating hedging strategies.

3.6.2 Episode Construction

The historical price series is divided into non-overlapping episodes of length 40 trading days. Each episode follows the same structure used in the synthetic experiments:

- The first 20 days are used to construct feature vectors describing recent market behaviour.
- Hedging decisions are applied over the remaining 20 days.

This procedure produces a dataset of hedging episodes extracted from the historical price series. Because the models were trained only on simulated Heston environments, this evaluation represents a true out-of-distribution test.

3.6.3 Evaluation Procedure

For each episode, the following hedging strategies are evaluated:

- Black–Scholes delta hedging with volatility estimated using an EWMA volatility estimator.
- Specialised deep hedging policies trained under the low-, medium-, and high-volatility simulation regimes.
- The regime-adaptive meta-model, which selects one of the specialised hedgers based on the observed market features at the start of the episode.

Two evaluation settings are considered:

- **Frictionless setting:** assumes no transaction costs.

- **Cost-aware setting:** incorporates transaction costs according to the specification introduced in Equation 2.15, with a proportional fee rate of 1% of traded notional and a minimum fee of USD 0.01 per trade.

Hedging performance is evaluated using the same replication error metrics defined in Section 3.5: RMSE, MAE, and mean hedging error.

3.6.4 Quantitative Results

Table 3.7 reports the aggregated hedging performance across all extracted episodes in the frictionless setting, while Table 3.8 reports the corresponding results when transaction costs are included.

Table 3.7: Real market hedging performance across all assets (frictionless setting).

Strategy	RMSE	MAE	Mean
BS_EWMA	0.01714	0.01202	-0.00093
Deep_low	0.01639	0.01169	-0.00208
Deep_med	**0.01543**	0.01158	-0.00051
Deep_high	0.01593	0.01197	-0.00021
Meta Model	0.01570	0.01139	-0.00110

Table 3.8: Real market hedging performance across all assets (with transaction costs).

Strategy	RMSE	MAE	Mean
BS_EWMA	0.02921	0.02545	-0.02165
Deep_low	0.02474	0.01935	-0.01453
Deep_med	**0.02086**	0.01763	-0.01319
Deep_high	0.02148	0.01794	-0.01337
Meta Model	0.02164	0.01776	-0.01339

In the frictionless setting, all learning-based strategies exhibit comparable performance on historical equity data, with only modest differences across metrics. The medium-volatility deep hedger achieves the lowest RMSE, while the meta-model attains the lowest MAE. This indicates that policies trained under simulated stochastic volatility environments are able to transfer reasonably well to real market dynamics, even without explicit calibration to historical data.

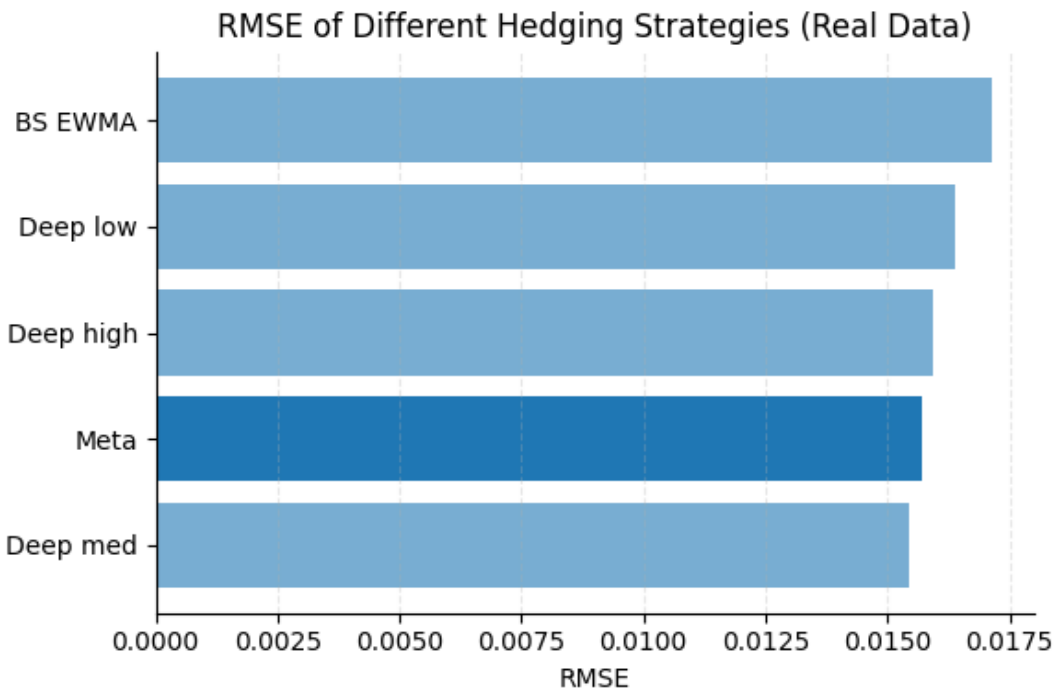


Figure 3.4: RMSE across episodes for different hedging strategies (Real Data)

When transaction costs are introduced, hedging performance deteriorates across all strategies, reflecting the impact of trading frictions on dynamic rebalancing. This deterioration is accompanied by more negative mean hedging errors, indicating a systematic reduction in terminal portfolio value due to trading costs.

In this setting, the medium-volatility deep hedger achieves the strongest overall performance, while the meta-model remains competitive and continues to outperform the Black–Scholes EWMA benchmark. These results suggest that learning-based hedging policies are better able to balance replication accuracy and trading costs in practice, leading to more stable performance under realistic market conditions.

The average absolute hedging error across episodes over time is visualised in Figure 3.6 for the frictionless setting and Figure 3.7 for the cost-aware setting.

These figures provide a time-series perspective on the performance of the evaluated hedging strategies.

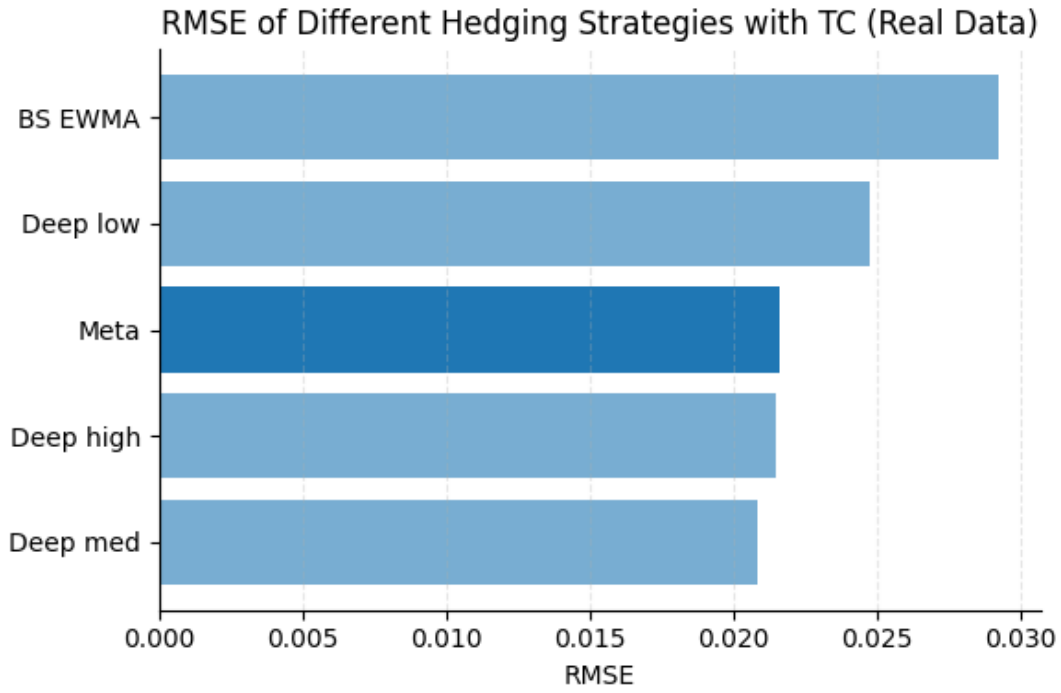


Figure 3.5: RMSE across episodes for different hedging strategies with transaction costs (Real Data)

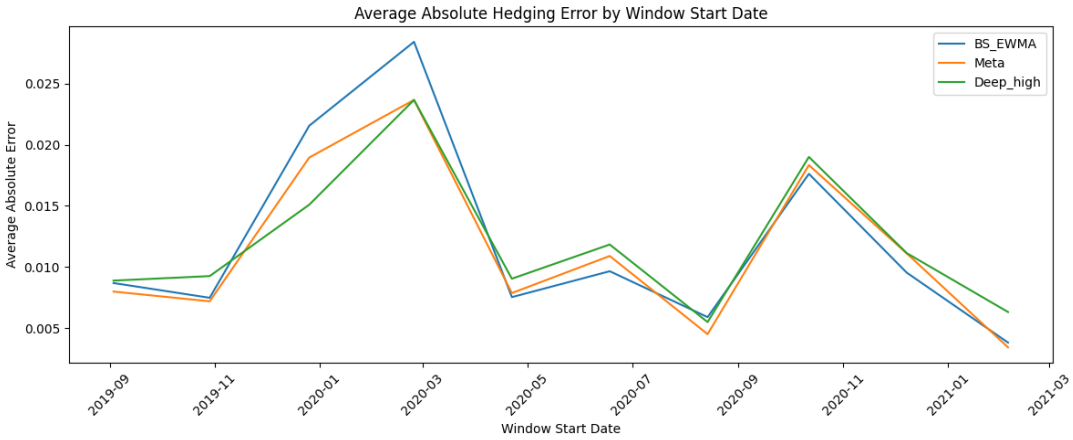


Figure 3.6: Average absolute hedging error across episodes for different hedging strategies over time in the frictionless setting.

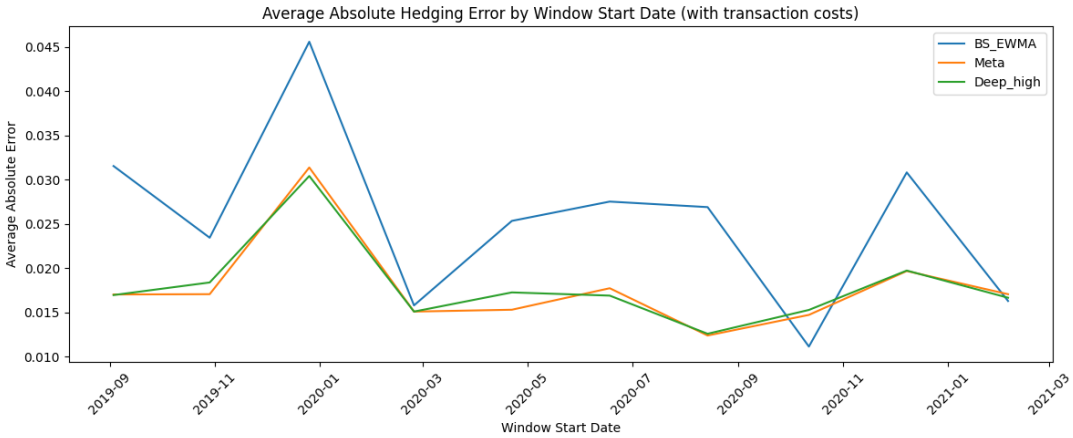


Figure 3.7: Average absolute hedging error across episodes for different hedging strategies over time in the cost-aware setting.

3.6.5 Market Volatility During the Sample Period

To visualise the changing market environment during the evaluation period, Figure 3.8 plots the realised volatility proxy computed using a 20-day rolling window.

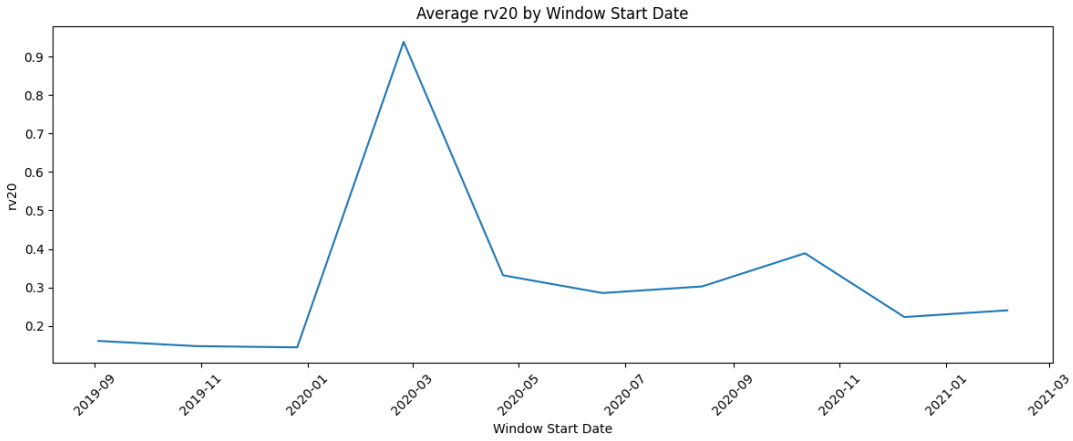


Figure 3.8: 20-day realised volatility (RV20) of the underlying assets during the evaluation period. The sharp increase in volatility during early 2020 corresponds to the COVID-19 market crash.

The figure illustrates the substantial increase in market volatility during early 2020, followed by a gradual stabilisation in later periods. This variation in market conditions provides a natural test of the robustness of the proposed regime-adaptive hedging framework.

3.6.6 Hedging Error Comparison

To compare the behaviour of different hedging strategies across time, Figure 3.9 plots the difference between the absolute hedging error of the meta model and the Black–Scholes EWMA strategy in the frictionless setting, while Figure 3.10 presents the same comparison in the cost-aware setting.

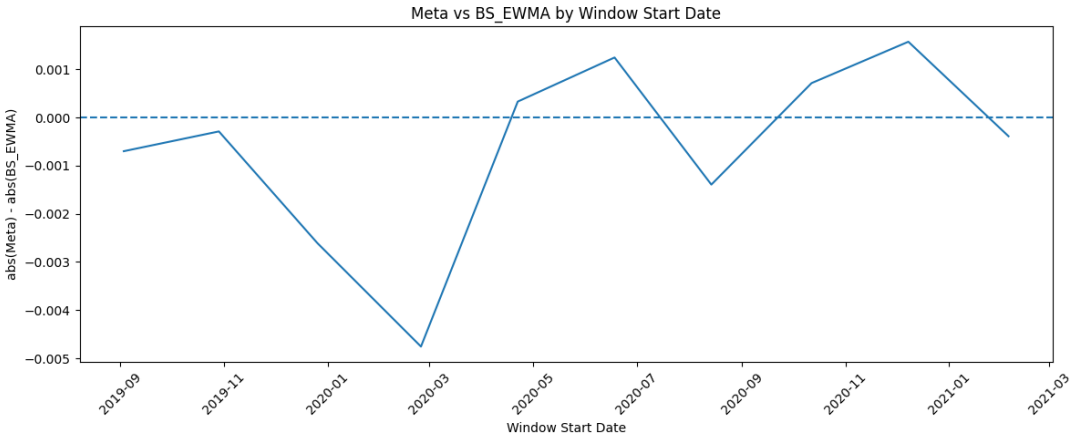


Figure 3.9: Difference in absolute hedging error between the meta model and the Black–Scholes EWMA strategy in the frictionless setting. Negative values indicate periods where the meta model achieves lower hedging error.

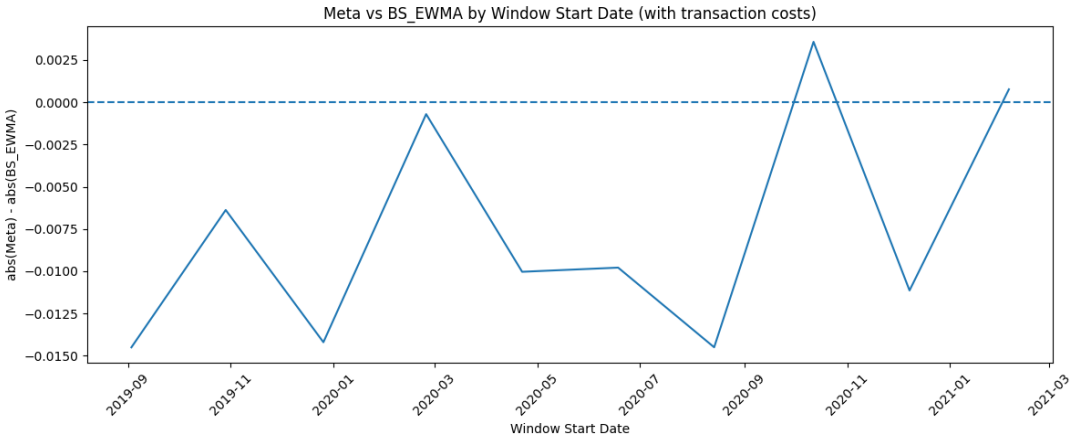


Figure 3.10: Difference in absolute hedging error between the meta model and the Black–Scholes EWMA strategy in the cost-aware setting. Negative values indicate periods where the meta model achieves lower hedging error.

The results suggest that the regime-adaptive approach remains competitive relative to the Black–Scholes baseline across time, particularly during more volatile periods when a single volatility estimate may be less reliable.

3.6.7 Temporal Performance During the COVID Period

To further analyse model behaviour across changing market conditions, hedging errors are examined during the COVID-19 volatility shock.

Figure 3.11 plots the average absolute hedging errors across episodes during the sample period in the frictionless setting, while Figure 3.12 shows the corresponding behaviour in the cost-aware setting.

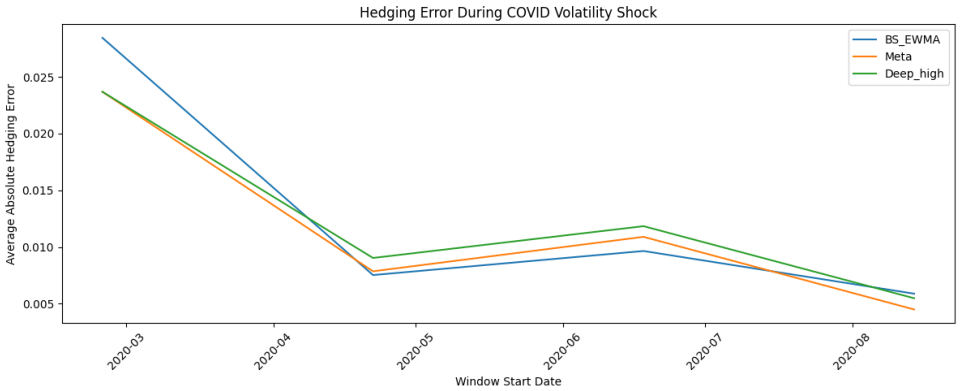


Figure 3.11: Average absolute hedging error across episodes during the COVID-19 market period in the frictionless setting.

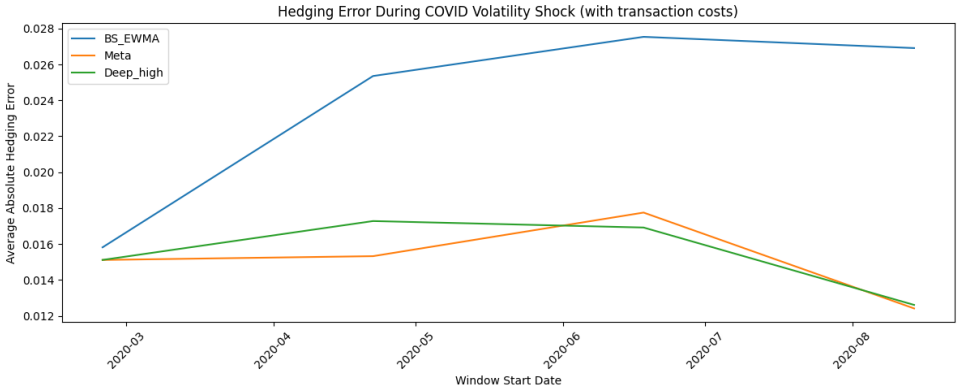


Figure 3.12: Average absolute hedging error across episodes during the COVID-19 market period in the cost-aware setting.

These figures illustrate how hedging performance evolves during periods of elevated market volatility. The regime-adaptive framework remains competitive across both settings, while the inclusion of transaction costs amplifies the penalty for frequent hedge adjustments during stressed market conditions.

A detailed interpretation of these results is provided in Chapter 4.

Chapter 4

Discussion and Limitations

This chapter interprets the experimental findings presented in Chapter 3. The results are analysed in two stages. First, the behaviour of the hedging strategies is examined in the synthetic Heston environment where the underlying volatility regime is controlled. Second, the robustness of the proposed framework is evaluated using real market data. The chapter concludes with a discussion of the limitations of the current approach.

4.1 Synthetic Market Results

The synthetic experiments provide a controlled environment for evaluating the behaviour of the regime-adaptive hedging framework. Because the volatility regimes are known by construction, the experiments allow direct comparison between specialised deep hedgers, classical Black–Scholes hedging, and the proposed meta-model.

The results demonstrate that specialised deep hedgers consistently achieve the lowest replication error when evaluated within the volatility regime under which they were trained. This confirms that the learning-based policies are able to internalise regime-specific hedging behaviour from the simulated environment. In particular, each model adapts its hedge adjustments to the statistical characteristics of its training regime, such as volatility persistence and variance dynamics.

When episodes across all regimes are pooled, the regime-adaptive meta-model achieves the strongest aggregate performance. By selecting among specialised hedgers based on observable features, the meta-model effectively approximates a regime-aware strategy

without direct access to the true underlying regime. This highlights a key practical advantage: while specialised models require prior knowledge of the regime, the meta-model operates under realistic conditions where regimes must be inferred from market data.

The inclusion of transaction costs introduces an additional dimension to this comparison. While all strategies experience increased hedging error under trading frictions, the relative performance changes. In particular, the advantage of the meta-model becomes less pronounced, and in some cases, a robust specialised hedger achieves the lowest error. This suggests that, in the presence of costs, consistently applying a stable hedging policy may be preferable to dynamically switching between regimes.

Overall, the synthetic experiments highlight a trade-off between adaptability and stability. Regime-adaptive strategies perform well in frictionless settings with heterogeneous environments, while more stable, specialised policies may be advantageous when trading costs are taken into account.

4.2 Real Market Results

The real market evaluation provides a more stringent test of the proposed framework, as the models are trained entirely on simulated Heston environments but evaluated on historical equity price data. This setting assesses the extent to which the learned hedging policies generalise beyond the training distribution.

In the frictionless setting, all learning-based strategies exhibit comparable performance, with only modest differences across metrics. This indicates that policies trained on simulated stochastic volatility environments are able to transfer reasonably well to real market dynamics, despite the presence of model misspecification.

The time-series analysis reveals that hedging performance varies significantly across different market conditions. During relatively stable periods, the Black–Scholes benchmark performs comparably to the deep hedging strategies, reflecting the suitability of its constant volatility assumption in low-volatility environments.

In contrast, during periods of heightened market volatility, such as the COVID-19 market shock, the deep hedging strategies demonstrate improved robustness. These models are better able to accommodate rapid changes in market conditions, likely due to their exposure to stochastic volatility dynamics during training.

When transaction costs are introduced, the differences between strategies become more pronounced. The Black–Scholes EWMA strategy deteriorates significantly, reflecting the cost of frequent hedge adjustments in response to short-term volatility fluctuations. In comparison, the deep hedging strategies exhibit smoother performance profiles, suggesting that they implicitly learn more cost-efficient hedging behaviour.

The meta-model remains competitive across both settings, although it does not consistently outperform the best specialised hedger. Instead, its primary strength lies in delivering stable performance across a wide range of market conditions. This aligns with the objective of the regime-adaptive framework, which is to provide robust hedging performance under uncertainty regarding the underlying volatility regime.

Taken together, these results suggest that the main advantage of learning-based hedging policies lies not only in improved replication accuracy, but also in their ability to balance responsiveness and trading intensity in realistic market environments.

4.3 Limitations

Despite the promising results, several limitations of the current study should be acknowledged.

First, the training environment is based on the Heston stochastic volatility model. While this model captures key stylised facts such as stochastic variance and volatility clustering, real financial markets may exhibit additional features such as jumps, structural breaks, and varying liquidity conditions. These effects are not captured in the current simulation framework.

Second, while transaction costs are incorporated in the extended experiments, the model assumes a simplified cost structure. In practice, trading costs may depend on ad-

ditional factors such as market liquidity, order size, and execution strategy. Incorporating more realistic microstructure effects would provide a more comprehensive evaluation.

Third, the regime classification component relies on a relatively simple feature set derived from recent returns and volatility estimates. More advanced approaches, such as representation learning or sequential models, could potentially improve regime identification and, consequently, the performance of the meta-model.

Fourth, the set of volatility regimes considered in the simulation is limited. Real markets may exhibit a richer and more continuous spectrum of regimes, and extending the framework to incorporate a larger and more flexible set of hedging policies may further enhance its adaptability.

Finally, the meta-classification model introduces an additional source of uncertainty through potential misclassification of market regimes. When the predicted regime does not align with the underlying market conditions, the selected hedging policy may be suboptimal, particularly during periods where regime boundaries are less distinct or evolve gradually over time. This reflects an inherent trade-off between adaptability and classification accuracy within the proposed framework.

These limitations suggest several directions for future work, including the incorporation of richer feature representations, more flexible regime definitions, and extensions to more realistic market settings.

Chapter 5

Conclusion and Future Work

This thesis studies the problem of dynamic option hedging under stochastic volatility. In practical financial markets, volatility evolves over time and is not directly observable, making perfect replication impossible using the underlying asset alone. As a result, hedging strategies must operate under both model uncertainty and changing market conditions.

5.1 Summary of Contributions

The primary objective of this thesis is to develop a hedging framework that remains robust across heterogeneous volatility regimes. To address this, a regime-adaptive deep hedging framework was proposed, combining specialised deep hedging policies with a meta-classification model that selects an appropriate policy based on observable market features.

The framework was evaluated in both simulated and real market environments. In synthetic experiments under the Heston stochastic volatility model, specialised deep hedgers were shown to learn regime-dependent hedging behaviour and achieve strong performance within their respective environments. When evaluated across mixed regimes, the meta-model achieved the best aggregate performance by adapting its policy selection to changing conditions.

The inclusion of transaction costs revealed an additional dimension of performance. While all strategies deteriorate under trading frictions, learning-based hedging policies

exhibit greater stability, and the relative advantage of dynamic regime switching becomes less pronounced. In such settings, robust specialised hedgers can perform comparably to, or even better than, the regime-adaptive approach.

When applied to real equity data spanning the COVID-19 volatility shock, the proposed framework demonstrated consistent and competitive performance despite being trained entirely on simulated data. This suggests that the learned policies capture structural features of hedging under stochastic volatility that generalise beyond the training environment.

Overall, the results demonstrate that combining specialised hedging policies with a regime-selection mechanism provides a practical and effective approach to improving hedging robustness under volatility uncertainty, particularly in the presence of model misspecification and trading frictions.

5.2 Implications

The findings highlight the importance of accounting for regime uncertainty in dynamic hedging problems. While specialised models perform well within known environments, real financial markets are characterised by continuous transitions between different volatility regimes.

The proposed framework demonstrates that learning-based methods can be used to navigate this uncertainty by adapting hedging behaviour based on observable market conditions. Rather than replacing classical models, these approaches complement traditional methods by providing flexibility in environments where model assumptions are violated.

An additional insight is the role of trading frictions in shaping hedging performance. The results suggest that learning-based policies implicitly balance hedging accuracy and trading intensity, leading to more stable outcomes when transaction costs are present. This highlights the importance of evaluating hedging strategies in realistic settings that account for market frictions.

5.3 Future Work

Several directions for future research naturally arise from this study.

First, the simulation environment can be extended to incorporate richer market dynamics. While the Heston model captures stochastic volatility, real markets may exhibit jumps, regime-switching behaviour, or structural breaks. Training hedging policies under more complex dynamics could improve robustness and realism.

Second, although transaction costs are incorporated in the evaluation, further work could integrate more realistic market microstructure effects, such as bid-ask spreads, liquidity constraints, and execution delays. Incorporating these factors directly into the training process may lead to more practically deployable hedging strategies.

Third, the regime classification component can be enhanced using more advanced representation learning techniques. Methods such as recurrent neural networks or attention-based architectures could better capture temporal dependencies and improve regime identification.

Finally, future work could explore extending the framework to a continuous set of hedging policies, allowing the meta-model to act as a soft allocation mechanism rather than a discrete selector. This would align the approach more closely with mixture-of-experts models and enable smoother adaptation to evolving market conditions.

In summary, this thesis demonstrates that regime-adaptive learning frameworks provide a principled and effective approach to improving the robustness of dynamic hedging strategies, particularly in environments characterised by stochastic volatility and market frictions. These results highlight the potential of combining financial modelling with machine learning to address challenges arising from model uncertainty and evolving market conditions.

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